

Nicolò Carpentieri

SYSTEMS & COMPUTER ARCHITECTURE

PhD Candidate, Systems Research Group
Technical University of Munich

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01 RESEARCH INTERESTS

My interests lie in **computer architecture and systems**, particularly **memory disaggregation** at rack scale and the scale-up interconnects, like CXL and related fabrics, that make it possible. Within that space, I focus on **cache coherence protocols** for heterogeneous accelerators, including GPUs, and on automating the design and verification of the controllers that implement them.

02 EDUCATION

Ph.D. in Computer Science Oct 2024 -- present
Technical University of Munich
Advisor: Prof. Dr. Pramod Bhatotia

M.Sc. in Computer Engineering 2021 -- 2024
Politecnico di Torino
Thesis: Designing and Evaluating Mapping of CNN Layers on an Edge-CGRA

B.Sc. in Electronic Engineering 2018 -- 2021
Politecnico di Torino

03 EXPERIENCE

RESEARCH

Ph.D. Researcher, Technical University of Munich Oct 2024 -- present
Group: Chair of Computer Systems · Host: Prof. Dr. Pramod Bhatotia

- **CXL memory consistency:** shared remote CXL memory across heterogeneous architectures.
- **CXL bridges:** integrating CXL 3.0 cache coherence into arbitrary host architectures.
- **CXL GPUs:** GPUs with CXL 3.0 integration for symmetric cache coherence with host CPUs.

UNDERGRADUATE RESEARCH PROJECT

Visiting Student, Embedded Systems Laboratory (ESL), EPFL 2023 -- 2024
Mapping of convolutional neural network algorithms onto CGRA architectures, in collaboration with DAUIN, Politecnico di Torino.

04 PUBLICATIONS

CONFERENCES

- [C1] Decoupling Protocol, Topology, and Policy for Automated Synthesis of Coherence Controllers
Nicolò Carpentieri, Anatole Lefort, David Schall, Julian Pritzi, Pramod Bhatotia
IEEE/ACM MICRO 2026
- [C2] vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures
Anatole Lefort*, Julian Pritzi*, Nicolò Carpentieri, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia (*equal contribution)
ACM ASPLOS 2026 ★ Best Paper Award
- [C3] C3: CXL Coherence Controllers for Heterogeneous Architectures
Anatole Lefort, David Schall, Nicolò Carpentieri, Julian Pritzi, Nicolai Oswald, Pramod Bhatotia

IEEE HPCA 2026

[C4] Performance Evaluation of Acceleration of Convolutional Layers on OpenEdgeCGRA

Nicolò Carpentieri, Juan Sapriz, Davide Schiavone, Daniele Jahier Pagliari, David Atienza, Maurizio Martina, Alessio Burrello

ACM CF 2024

JOURNALS

[J1] Invited extended journal version of vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures

Anatole Lefort*, Julian Pritzi*, **Nicolò Carpentieri**, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia (*equal contribution)

ACM Transactions on Computer Systems (TOCS) forthcoming, 2026

05 TALKS

CONFERENCES

Decoupling Protocol, Topology, and Policy for Automated Synthesis of Coherence Controllers

IEEE/ACM MICRO 2026, Athens, Greece

2026

WORKSHOPS

ProtoSLICC: Automated Synthesis of Gem5-based Cache Coherence Controllers

gem5 Workshop @ ISCA 2025, Tokyo, Japan

2025

06 ACADEMIC SERVICE

Artifact Evaluation Committee

2026

ACM ASPLOS 2026

IEEE/ACM MICRO 2026

07 STUDENT VOLUNTEERING

Member, IEEE-HKN (Mu Nu Chapter)

2022 -- 2024

Politecnico di Torino

08 TEACHING EXPERIENCE

Introduction to Software Engineering (EIST) — *Examiner*

2024--2026

Systems Programming — *Course Organizer*

2026--2027

09 STUDENT SUPERVISION

Disaggregated, Vendor-Generic APUs with CXL

Victor Fritz Trost — B.Sc. Thesis

SS 2025

10 HONORS AND AWARDS

Induction in IEEE-HKN (Mu Nu Chapter), Politecnico di Torino chapter

Jun 2022

11 SKILLS

Programming languages C/C++ · Verilog HDL · VHDL · Python · Shell script · $\LaTeX$ · Matlab

Development Unix/Linux · gem5 simulator · FPGA · CUDA · RDMA

12 OTHER INTERESTS

Sports Skiing · Tennis · Football · Basketball · Padel · Volleyball

Music Piano · Flute (8+ years)